

Ahmed Shaaban

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Location : Muscat, Oman

LANGUAGES AND TOOLS

- Python, JavaScript, Java
- LangChain, DBT, PostgreSQL, MetaBase, DVC, Docker, FastAPI, TensorFlow, Keras, KerasOcr, Pytorch, FastAI, mySQL, openCV, Tableau, React Native, React.js, Next.js, Node.js, Jinja, Label studio, CVAT.
- **Source Control:** Git
- Development Methodologies:** Agile

PREVIOUS EXPERIENCE

- **Esbaar** Muscat, Oman
Senior AI Engineer Oct 2024 - Present
 - **Esbaar ChatBot:**
 - Developed internal chatbot to assist employees with company regulations using **LLMs** and **RAG** methods.
 - Impact:**
 - Reduced HR requests by **30%**.
 - **Rig AI:**
 - Developed real-time video analysis system using **GStreamer** and **DeepStream**.
 - Optimized production models using **Quantized-aware training** and **Post-training calibration**.
 - Applied system analysis using **PostgreSQL** and **MetaBase**.
 - Impact:**
 - Increased the system's ability to process **real-time** cameras' feed from **3** to **14**.
 - **Esbaar Data Pipeline:**
 - Developed a data sampler to extract most relevant frames using **Tensorflow** and **OpenCV**.
 - Developed annotation generator for **Active learning** using **PyTorch**, **Yolo** and **OpenCV**.
 - Developed annotation validator to extract anomalies in labels using **PyTorch** and **OpenCV**.
 - Managed the entire pipeline and data storage using **DVC**.
 - Impact:**
 - Reduced redundant frames by **96%**.
 - Reduced data annotation time by **66%**.
 - Reduced overall time to release new models by **50%**.
- **Treyd** Stockholm, Sweden
Full Stack Data Scientist Oct 2021 - Oct 2024
 - **Invoice AI:**
 - Developed text classification and recommendation system using **LLMs**, **Google's OCR** and **OpenCV**.
 - **Fine-tuned** the **LLM** to enhance performance and ensure consistent responses.
 - Developed deep learning based invoice fields detection system using **PyTorch**.
 - Deployed the backend using **fastAPI** and **Docker** and integrate it with **Google's CLOUD RUN**.
 - Impact:**
 - Automated The approval for **40%** of invoices.
 - Reduced invoice approval time for the other **60%** of invoices from **30** minutes to **2** minutes.
 - **Companies Behaviour Analysis:**
 - Measure data quality of companies' payable and receivables invoices data
 - Analyze companies' financial decisions based on their payable and receivable data.
 - Measure concentration risk based on how much companies relies on its suppliers and customers.
 - Impact:**
 - Improved detection of financially unstable companies by **20%**.
 - **Treyd Data Warehouse:**
 - Designed data warehouse using **DBT** and **PostgreSQL** for decision making with analytics using **MetaBase**.
 - Impact:**
 - Sales team **doubled** the number of active customers after using the customer-related analytics.
 - Finance team improve the quality of the company's portfolio by **48%**.
 - Marketing team increased the number of partnerships by **20%**.
 - Improved query execution time by **3x** due to moving from **OLTP** to **OLAP** design principles.
 - Improved the quality of reporting due to the use of **OBT** design.

- **Blink consumer finance** Cairo, Egypt
Full Stack Data Scientist *June 2020 - July 2021*
 - **EKYC system:**
 - Developed Arabic custom **OCR** using **LSTMs** with tools like **TensorFlow** and **Keras**.
 - Built Mobile Real-time client-side **OCR** for Arabic numbers and dates using **OpenCV**.
 - Built Image manipulation detector using **Local Binary Patterns**.
 - Built real-time **facial detection, recognition and tracking** modules using **Google** and **AWS** facial systems.
 - Built multi **card detector** on complicated and multi-colored backgrounds using **OpenCV**.
 - Built general **document classification** model using **TensorFlow** and **Kears**.
 - Built mobile real-time client-side **card scanner** using **OpenCV**.
 - Integrated The previous computer vision algorithms with cross platform mobile app using **React Native**.
 - Impact:**
 - Reduced average customer onboarding time from **72 hours** to **3 minutes**.
- **Advintic** Leipzig, Germany
Machine Learning Engineer *Jan 2020 - June 2020*
 - **Main heart arteries detector :**
 - Built data preprocessing pipeline to process Dicom CT images.
 - Modified the RESNET architecture layers to be able to handle regression localization problems.
 - Developed vision-based system to identify coronary artery regions in CT scans, aiding diagnosis, using **fastAI** and **PyTorch**.
 - Impact:**
 - Increased overall detection accuracy from **87%** to **94%**.
- **Systems Lab** Waterloo University, Canada
Machine Learning Researcher (Remote) *Jan 2019 - June 2020*
 - **Robotic Hand Controller:**
 - Developed data preprocessing pipeline to modify and augment the data.
 - Developed vision-based system to detect hand grasp types for amputees using **ResNet**, **PyTorch** and **FastAI**.
 - Impact:**
 - Increased overall detection accuracy from **84%** to **97%**.

EDUCATION

- **Cairo University** Egypt
Bachelor of Science: Systems Engineering;
 - **Undergraduate Coursework:** Machine Learning, Computer Vision, Probability and Statistics, Calculus and Linear Algebra, Digital Signal Processing, Data Structures and Algorithms, Databases, Computer Networks, Micro controllers, Computer Graphics, Robotics.

CERTIFICATES

- **Data Science Track:** 365 Data Science [certificate]
- **NLP Specialization:** Coursera [certificate]
- **Deep Learning Specialization:** Coursera [certificate]
- **Data Analysis Track:** Udacity [certificate]

BOOKS

- **Building LLMs For Production.**
- **Elements Of Statistical Learning.**
- **Machine Learning Probabilistic Perspective.**
- **The Manager's Path.**
- **Kimball The Data Warehouse Toolkit 3rd-Edition.**